

A 3D Nonconforming Finite Element Method for Hodge-Laplace Problem

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March 19, 2026

1 Problem

1.1 Strong Formulation

Let $V = H_0(\text{curl}; \Omega) \cap H(\text{div}; \Omega)$

Find $\lambda \in \mathbb{R}$, $u \in V$, such that

$$\begin{cases} \text{curl curl } u - \text{grad div } u = \lambda u, & \text{in } \Omega \\ u \times n = 0, & \text{on } \partial\Omega \\ \text{div } u = 0, & \text{on } \partial\Omega \end{cases} \quad (1)$$

1.2 Primal Weak Formulation

Find $\lambda \in \mathbb{R}$, $u \in V$, such that

$$(\text{div } u, \text{div } v) + (\text{curl } u, \text{curl } v) = \lambda(u, v), \quad \forall v \in V \quad (2)$$

1.3 Mixed Formulation

Find $\lambda \in \mathbb{R}$, $\sigma \in H(\text{curl}; \Omega)$, $u \in H(\text{div}; \Omega)$, such that

$$\begin{cases} (\sigma, \tau) - (u, \text{curl } \tau) = 0, \forall \tau \in H(\text{curl}; \Omega) \\ (\text{curl } \sigma, v) + (\text{div } u, \text{div } v) = \lambda(u, v), \forall v \in H(\text{div}; \Omega) \end{cases} \quad (3)$$

2 Finite Element Space

We define

$$P_{rd}(T) = \text{span}\left\{ \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}, \begin{pmatrix} \tilde{x} \\ \tilde{y} \\ \tilde{z} \end{pmatrix}, \begin{pmatrix} 0 \\ -\tilde{z} \\ \tilde{y} \end{pmatrix}, \begin{pmatrix} \tilde{z} \\ 0 \\ -\tilde{x} \end{pmatrix}, \begin{pmatrix} -\tilde{y} \\ \tilde{x} \\ 0 \end{pmatrix}, \right. \\ \left. \begin{pmatrix} 2\tilde{x}^2 - (\tilde{y}^2 + \tilde{z}^2) \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 2\tilde{y}^2 - (\tilde{x}^2 + \tilde{z}^2) \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \\ 2\tilde{z}^2 - (\tilde{x}^2 + \tilde{y}^2) \end{pmatrix} \right\} \quad (4)$$

The Finite Element Space is defined as follows:

$$V_h := \{\sigma_h \in P_{rd}(\mathcal{G}_h) \mid (\operatorname{div}_h \sigma_h, \psi_h) + (\sigma_h, \nabla_h \psi_h) = 0, \forall \psi_h \in V_{h0}^{CR},$$

$$\text{and } (\operatorname{curl}_h \sigma_h, \tau_h) - (\sigma_h, \operatorname{curl}_h \tau_h) = 0, \forall \tau_h \in V_N\} \quad (5)$$

2.1 Local Basis Function

Let T be a tetrahedral, $\psi_i (i = 1 : 4)$ are *CR* elements and $\tau_i (i = 1 : 6)$ are *Nedelec* elements, our goal is to find $\tilde{\omega}_i (i = 1 : 10) \in P_{rd}(T)$, such that

$$\int_T \operatorname{div} \tilde{\omega}_i \cdot \psi_j + \tilde{\omega}_i \cdot \nabla \psi_j = \delta_{ij}, i = 1 : 10, j = 1 : 4 \quad (6)$$

$$\int_T \operatorname{rot} \tilde{\omega}_i \cdot \tau_j - \tilde{\omega}_i \cdot \operatorname{curl} \tau_j = \delta_{i-4,j}, i = 1 : 10, j = 1 : 6 \quad (7)$$

First, we define two function space:

$$\omega_1, \dots, \omega_{10} := \left\{ \begin{pmatrix} \tilde{x} \\ \tilde{y} \\ \tilde{z} \end{pmatrix}, \begin{pmatrix} 0 \\ -\tilde{z} \\ \tilde{y} \end{pmatrix}, \begin{pmatrix} \tilde{z} \\ 0 \\ -\tilde{x} \end{pmatrix}, \begin{pmatrix} -\tilde{y} \\ \tilde{x} \\ 0 \end{pmatrix}, \begin{pmatrix} 2\tilde{x}^2 - (\tilde{y}^2 + \tilde{z}^2) + \frac{(B+C-2A)}{V} \\ 0 \\ 0 \end{pmatrix}, \right.$$

$$\left. \begin{pmatrix} 0 \\ 2\tilde{y}^2 - (\tilde{x}^2 + \tilde{z}^2) + \frac{(A+C-2B)}{V} \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \\ 2\tilde{z}^2 - (\tilde{x}^2 + \tilde{y}^2) + \frac{(A+B-2C)}{V} \end{pmatrix}, \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \right\}$$

$$(\psi, \eta)_1, \dots, (\psi, \eta)_{10} := \left\{ (1, \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}), (0, \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}), (0, \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}), (0, \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}), (2\tilde{x}, \begin{pmatrix} 0 \\ -\tilde{z} \\ \tilde{y} \end{pmatrix}), (2\tilde{y}, \begin{pmatrix} \tilde{z} \\ 0 \\ -\tilde{x} \end{pmatrix}), \right.$$

$$\left. (2\tilde{z}, \begin{pmatrix} -\tilde{y} \\ \tilde{x} \\ 0 \end{pmatrix}), (2\tilde{x}, \begin{pmatrix} 0 \\ \tilde{z} \\ -\tilde{y} \end{pmatrix}), (2\tilde{y}, \begin{pmatrix} -\tilde{z} \\ 0 \\ \tilde{x} \end{pmatrix}), (2\tilde{z}, \begin{pmatrix} \tilde{y} \\ -\tilde{x} \\ 0 \end{pmatrix}) \right\}$$

and two operator:

$$T\omega := (\operatorname{div} \omega, \operatorname{rot} \omega) \quad (8)$$

$$T^*(\psi, \eta) := \nabla \psi - \operatorname{curl} \eta \quad (9)$$

We have a bilinear form:

$$a(\omega, (\psi, \eta)) := (T\omega, (\psi, \eta)) + (\omega, T^*(\psi, \eta)) \quad (10)$$

compute the inner product matrix:

$$L_1 = \begin{bmatrix} 3V & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 2V & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 2V & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 2V & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 8A+2B+2C & 6D & 6E & 8A-2B-2C & 10D & 10E & 10F \\ 0 & 0 & 0 & 0 & 6D & 8B+2A+2C & 6F & 10D & 8B-2A-2C & 10F & 8C-2A-2B \\ 0 & 0 & 0 & 0 & 6E & 6F & 8C+2A+2B & 10E & 10F & 8C-2A-2B & 4V \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 4V & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 4V & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 4V & 0 \end{bmatrix}$$

and the inverse matrix:

$$L_2 = \begin{bmatrix} \frac{1}{3V} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{1}{2V} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{1}{2V} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{1}{2V} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & a_{11} & a_{12} & a_{13} & Ma_{11} - \frac{5}{12V} & Ma_{12} & Ma_{13} \\ 0 & 0 & 0 & 0 & a_{21} & a_{22} & a_{23} & Ma_{21} & Ma_{22} - \frac{5}{12V} & Ma_{23} \\ 0 & 0 & 0 & 0 & a_{31} & a_{32} & a_{33} & Ma_{31} & Ma_{32} & Ma_{33} - \frac{5}{12V} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{4V} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{4V} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{4V} \end{bmatrix}$$

where $a_{11} = \frac{(A+4B+C)(A+B+4C)-9F^2}{K}$, $a_{12} = \frac{-3(A+B+4C)D+9EF}{K}$, $a_{13} = \frac{-3(A+4B+C)E+9DF}{K}$,
 $a_{21} = a_{12}$, $a_{22} = \frac{(4A+B+C)(A+B+4C)-9E^2}{K}$, $a_{23} = \frac{-3(4A+B+C)E+9DF}{K}$, $a_{31} = a_{13}$, $a_{32} = a_{23}$,
 $a_{33} = \frac{(4A+B+C)(A+4B+C)-9D^2}{K}$
 $M = \frac{4(A+B+C)}{3V}$

$$K = 8(A+B+C)^3 + 18[(A((B+C)^2 - (D^2 + E^2 + 4F^2)) + B((A+C)^2 - (D^2 + 4E^2 + F^2)) + C((A+B)^2 - (4D^2 + E^2 + F^2)) + 6DEF]$$

The second step, we need to use $(CR, Nedelec)$ to represent (ψ, η) , the represent matrix is:

$$L_3 = \begin{bmatrix} 1 & 1 & 1 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \tilde{x}_2 - \tilde{x}_1 & \tilde{x}_3 - \tilde{x}_1 & \tilde{x}_4 - \tilde{x}_1 & \tilde{x}_3 - \tilde{x}_2 & \tilde{x}_4 - \tilde{x}_2 & \tilde{x}_4 - \tilde{x}_3 \\ 0 & 0 & 0 & 0 & \tilde{y}_2 - \tilde{y}_1 & \dots & \dots & \dots & \dots & \dots \\ 0 & 0 & 0 & 0 & \tilde{z}_2 - \tilde{z}_1 & \dots & \dots & \dots & \dots & \dots \\ -\frac{2}{3}\tilde{x}_1 & -\frac{2}{3}\tilde{x}_2 & -\frac{2}{3}\tilde{x}_3 & -\frac{2}{3}\tilde{x}_4 & \tilde{y}_1\tilde{z}_2 - \tilde{y}_2\tilde{z}_1 & \dots & \dots & \dots & \dots & \dots \\ -\frac{2}{3}\tilde{y}_1 & -\frac{2}{3}\tilde{y}_2 & -\frac{2}{3}\tilde{y}_3 & -\frac{2}{3}\tilde{y}_4 & \tilde{z}_1\tilde{x}_2 - \tilde{z}_2\tilde{x}_1 & \dots & \dots & \dots & \dots & \dots \\ -\frac{2}{3}\tilde{z}_1 & -\frac{2}{3}\tilde{z}_2 & -\frac{2}{3}\tilde{z}_3 & -\frac{2}{3}\tilde{z}_4 & \tilde{x}_1\tilde{y}_2 - \tilde{x}_2\tilde{y}_1 & \dots & \dots & \dots & \dots & \dots \\ -\frac{2}{3}\tilde{x}_1 & -\frac{2}{3}\tilde{x}_2 & -\frac{2}{3}\tilde{x}_3 & -\frac{2}{3}\tilde{x}_4 & \tilde{z}_1\tilde{y}_2 - \tilde{z}_2\tilde{y}_1 & \dots & \dots & \dots & \dots & \dots \\ -\frac{2}{3}\tilde{y}_1 & -\frac{2}{3}\tilde{y}_2 & -\frac{2}{3}\tilde{y}_3 & -\frac{2}{3}\tilde{y}_4 & \tilde{x}_1\tilde{z}_2 - \tilde{x}_2\tilde{z}_1 & \dots & \dots & \dots & \dots & \dots \\ -\frac{2}{3}\tilde{z}_1 & -\frac{2}{3}\tilde{z}_2 & -\frac{2}{3}\tilde{z}_3 & -\frac{2}{3}\tilde{z}_4 & \tilde{y}_1\tilde{x}_2 - \tilde{y}_2\tilde{x}_1 & \dots & \dots & \dots & \dots & \dots \end{bmatrix}$$

Finally, we get the basis:

$$\begin{pmatrix} \tilde{\omega}_1 \\ \tilde{\omega}_2 \\ \dots \\ \tilde{\omega}_{10} \end{pmatrix} = L_3' L_2 \begin{pmatrix} \omega_1 \\ \omega_2 \\ \dots \\ \omega_{10} \end{pmatrix}$$

$$\begin{aligned} \tilde{\omega}_1 = \frac{1}{3V}\omega_1 - \frac{2}{3}(\tilde{x}_1 a_{11} + \tilde{y}_1 a_{21} + \tilde{z}_1 a_{31})\omega_5 - \frac{2}{3}(\tilde{x}_1 a_{12} + \tilde{y}_1 a_{22} + \tilde{z}_1 a_{32})\omega_6 - \frac{2}{3}(\tilde{x}_1 a_{13} + \tilde{y}_1 a_{23} + \tilde{z}_1 a_{33})\omega_7 \\ - \frac{2}{3}M \begin{pmatrix} \tilde{x}_1 a_{11} + \tilde{y}_1 a_{21} + \tilde{z}_1 a_{31} \\ \tilde{x}_1 a_{12} + \tilde{y}_1 a_{22} + \tilde{z}_1 a_{32} \\ \tilde{x}_1 a_{13} + \tilde{y}_1 a_{23} + \tilde{z}_1 a_{33} \end{pmatrix} + \frac{1}{9V} \begin{pmatrix} \tilde{x}_1 \\ \tilde{y}_1 \\ \tilde{z}_1 \end{pmatrix} \end{aligned} \quad (11)$$

$\tilde{\omega}_i (i = 2, 3, 4)$ is similar to $\tilde{\omega}_1$.

$$\begin{aligned}
\tilde{\omega}_5 = & \frac{1}{2V}(\tilde{x}_2 - \tilde{x}_1)\omega_2 + \frac{1}{2V}(\tilde{y}_2 - \tilde{y}_1)\omega_3 + \frac{1}{2V}(\tilde{z}_2 - \tilde{z}_1)\omega_4 \\
& + ((\tilde{y}_1\tilde{z}_2 - \tilde{y}_2\tilde{z}_1)a_{11} + (\tilde{z}_1\tilde{x}_2 - \tilde{z}_2\tilde{x}_1)a_{21} + (\tilde{x}_1\tilde{y}_2 - \tilde{x}_2\tilde{y}_1)a_{31})\omega_5 \\
& + ((\tilde{y}_1\tilde{z}_2 - \tilde{y}_2\tilde{z}_1)a_{21} + (\tilde{z}_1\tilde{x}_2 - \tilde{z}_2\tilde{x}_1)a_{22} + (\tilde{x}_1\tilde{y}_2 - \tilde{x}_2\tilde{y}_1)a_{32})\omega_6 \\
& + ((\tilde{y}_1\tilde{z}_2 - \tilde{y}_2\tilde{z}_1)a_{13} + (\tilde{z}_1\tilde{x}_2 - \tilde{z}_2\tilde{x}_1)a_{23} + (\tilde{x}_1\tilde{y}_2 - \tilde{x}_2\tilde{y}_1)a_{33})\omega_7 - \frac{2}{3V} \begin{pmatrix} \tilde{y}_1\tilde{z}_2 - \tilde{y}_2\tilde{z}_1 \\ \tilde{z}_1\tilde{x}_2 - \tilde{z}_2\tilde{x}_1 \\ \tilde{x}_1\tilde{y}_2 - \tilde{x}_2\tilde{y}_1 \end{pmatrix} \quad (12)
\end{aligned}$$

$\tilde{\omega}_i (i = 6 : 10)$ is similar to $\tilde{\omega}_5$.

3 3D FEM Mesh

We use general tetrahedral mesh, define following matrices to store grid information:

Matrix	nodes	elements	element_edges	element_faces	N	faces_edges	global_edges	global_faces
Information	Point	P Number	E Number	S Number	Logical	E Number	P Number	P Number
Size	(np,3)	(nt,4)	(nt,6)	(nt,4)	(ns,2)	(ns,3)	(ne,2)	(ns,3)

Table 1: Matrix for Mesh

4 Freedom

Each face has four degrees of freedom, with the first one dual to the CR element and the remaining three dual to the Nedelec element.

However, it should be noted that the degrees of freedom processed in this way contain linearly dependent basis functions. The method to eliminate the correlation is to traverse each edge and delete a certain degree of freedom in the patch of this edge.

5 Stiffness Matrix

We need to compute two stiffness matrix: $(\text{div } \sigma_i, \text{div } \sigma_j)$ and $(\text{curl } \sigma_i, \text{curl } \sigma_j)$, and mass matrix (σ_i, σ_j) , on each tetrahedron, there are locally 16 basis functions, so the element stiffness matrix has 256 elements that need to be calculated.

6 Numerical Result

6.1 $\Omega = [0, 1]^3$

We contrast new nonconforming element and mixed element in the same mesh. (In fact, the real eigenvalue are $k\pi^2 (k = 2, 2, 2, 3, 3, 3, 4, 4, 4, 4, \dots)$)

6.1.1 New element

6.1.2 Mixed Element

We use *Nedelec* Element for $H(\text{curl}; \Omega)$ space and *RT* Element for $H(\text{div}; \Omega)$ space.

1	19.227	20.050	20.050	24.908	28.685	28.685	30.831	33.824	33.824	36.089
2	19.669	19.840	19.840	28.233	29.699	29.699	45.836	46.436	46.436	48.088
3	19.725	19.766	19.766	29.249	29.653	29.653	48.492	48.639	48.639	49.070
4	19.736	19.746	19.746	29.518	29.621	29.621	49.134	49.170	49.170	49.282

Table 2: New Element, Cube Domain

6.2 $\Omega = [0, 1]^3 \setminus (0.25, 0.5)^3$

6.3 $\Omega = [0, 1]^3 \setminus ((0.5, 0.75)^3 \cup (0, 25, 0.375)^2 \times [0, 1])$

6.4 $\Omega = [0, 1]^3 \setminus ((0.125, 0.25)^3 \cup (0.125, 0.25)^2 \times (0.5, 0.625) \cup (0.5, 0.625)^2 \times (0.375, 0.5))$